

Grade 11 Mathematics

Equations and Inequalities & Exponents and Surds

Task: Task

Term: Term 2

Time: 90 minutes

Total Marks: 50

QUESTION 1

1.1 Solve for x: $x^2 = 5x - 6$ (2)

1.2 Solve for x (correct to TWO decimal places): $4x - 3 = \frac{2}{x}$ (5)

1.3 Solve for x in each of the following, showing critical values clearly:

1.3.1 Solve for x: $x^2 - 4x - 21 \leq 0$ (3)

1.3.2 Solve for x: $x^2 + 6x + 9 > 0$ (2)

[12]

QUESTION 2

2.1 Simplify fully, showing all working: $2\sqrt{12} - 3\sqrt{27} + 5\sqrt{48}$ (4)

2.2 Solve for x: $2^{2x+1} - 9 \cdot 2^x + 4 = 0$ (5)

2.3 Solve for x in each of the following equations:

2.3.1 $2^{x+3} - 2^{x+1} = 96$ (4)

2.3.2 $3^{x+2} - 3^{x+1} - 3^{x-1} = 153$ (2)

[15]

TOTAL [50]

MARKING GUIDELINE

QUESTION 1		
1.1	$x^2 - 5x + 6 = 0 \rightarrow (x-2)(x-3) = 0$	✓ rearrangement and factorisation
	$x = 2$ or $x = 3$	✓ both values (2)
1.2	$4x^2 - 3x - 2 = 0$	✓ clear fraction and write in standard form
	$x = \frac{3 \pm \sqrt{(-3)^2 - 4(4)(-2)}}{2(4)}$	✓ substitution into the quadratic formula
	$x = \frac{3 \pm \sqrt{41}}{8}$	✓ simplify the discriminant
	$x = 1,18$ or $x = -0,43$	✓ ✓ both answers (5)
1.3.1	$(x-7)(x+3) \leq 0$	✓ factors and critical values
	$-3 \leq x \leq 7$	✓ ✓ both bounds (3)

1.3.2	$(x+3)^2 > 0$	✓ factors
	$x \in \mathbb{R}, x \neq -3$	✓ answer (both criteria) (2)

QUESTION 2		
2.1	$2\sqrt{12} = 2\sqrt{4 \cdot 3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$	✓ simplify first surd
	$3\sqrt{27} = 3\sqrt{9 \cdot 3} = 3 \cdot 3\sqrt{3} = 9\sqrt{3}$	✓ simplify second surd
	$5\sqrt{48} = 5\sqrt{16 \cdot 3} = 5 \cdot 4\sqrt{3} = 20\sqrt{3}$	✓ simplify third surd
	$4\sqrt{3} - 9\sqrt{3} + 20\sqrt{3} = 15\sqrt{3}$	✓ combine like surds (4)
2.2	Let $k = 2^x$, so $2^{2x+1} = 2 \cdot (2^x)^2 = 2k^2$, giving $2k^2 - 9k + 4 = 0$	✓ substitution to reveal quadratic
	$(2k-1)(k-4) = 0$	✓ factorise
	$k = \frac{1}{2}$ or $k = 4$	✓ values of k
	$2^x = \frac{1}{2}$ gives $x = -1$, and $2^x = 4$ gives $x = 2$	✓ ✓ both answers (5)
2.3.1	$2^{x+1}(2^2 - 1) = 96$	✓ factorise out lowest

		power
	$2^{x+1} \cdot 3 = 96$ so $2^{x+1} = 32$	✓ simplify coefficient
	$32 = 2^5$	✓ rewrite as same base
	$x+1=5$, so $x=4$	✓ solve for x (4)
2.3.2	$3^{x-1}(3^3 - 3^2 - 1) = 153$ so $3^{x-1} \cdot 17 = 153$	✓ factorise and simplify
	$3^{x-1} = 9 = 3^2$, so $x=3$	✓ solve for x (2)

TOTAL [50]

COGNITIVE LEVEL ANALYSIS GRID

QUESTION	Knowledge	Routine	Complex	Problem-Solving	Total
QUESTION 1	0	5	7	0	12
QUESTION 2	0	8	7	0	15
TOTAL	0	13	14	0	27
FET TARGET %	20%	35%	30%	15%	—
THIS TASK %	0%	26%	28%	0%	100%

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1+i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x-a)^2 + (y-b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$